

TANISH MISRA

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PROFILE

Sophomore CS student at Purdue applying ML and software engineering to production systems with real constraints, having experience across data pipelines, agentic tools, and embedded software, with a passion for Formula 1 and photography.

EDUCATION

Purdue University

West Lafayette, IN

B.S. Computer Science (Machine Intelligence & Security Concentrations), Statistics Minor Aug. 2025 – Dec. 2028

- Coursework: Problem Solving & OOP, Programming in C, The Data Mine, Linear Algebra, Discrete Mathematics
- Activities: Claude Builder Club, Google Developer Group, Purdue Electric Racing, Electric Vehicle Club, F1 @ Purdue

EXPERIENCE

Toyota Connected Experiences Division (CXD)

Plano, TX

Automation Co-op

May 2026 – Present

- Automated in-vehicle infotainment (IVI) test-case generation across 1,000+ manual procedures via a Python LLM pipeline, targeting an 80% cycle time reduction using schema-valid JSON and Pydantic v2.
- Built a swappable backend across 3 LLM providers (Azure, Vertex, Ollama), implementing retrieval-augmented prompting; scaled throughput via thread-pool concurrency, supported by a suite of 350+ tests and 2 CI pipelines.

The Data Mine, Purdue University

West Lafayette, IN

Undergraduate Data Science Researcher

Jan. 2026 – Present

- Completed 12 projects performing Exploratory Data Analysis (EDA), data wrangling, and geospatial visualization using pandas, Polars, NumPy, seaborn, GeoPandas, and Folium. Accepted into Corporate Partners for Fall 2026.

Q Spark Group (QSG)

Remote

Data, Analytics, and AI Intern

Mar. 2024 – Jul. 2025

- Led a PII privacy initiative training Transformer NER models on 1M+ synthetic records, improving recall to 90%.
- Analyzed 250,000+ SQL records via DataFrames, performing metadata analysis to organize company-wide data access.

Purdue Motorsport Student Organizations (PER & EVC)

West Lafayette, IN

Software Developer

Aug. 2025 – Present

- **Purdue Electric Racing Software Team:** Developing embedded C firmware for STM32 nodes with git-hash version mismatch detection for electric racing vehicles.
- **Electric Vehicle Club Autonomous Team:** Engineered a Dockerized OpenCV computer vision pipeline to compute real-time steering angles for an autonomous kart.

PROJECTS

Candle for Choroideremia (CHM) | Python, FastAPI, PostgreSQL, pgvector, OpenAI, RAG

Apr. 2026

- Created a research dashboard for CHM patients aggregating 29 trials and 475 publications, adopted at candleforchm.org after UAT by 3 blind patients. Featured in a CureCHM press release.
- Built on PostgreSQL/pgvector with FastAPI, a RAG pipeline, medical safety guardrails, and WCAG-compliant UI.

Securify | Python, LangGraph, RoBERTa, OpenAI | AlgoFest 2026 Hackathon Semi-Finalist

Apr. 2026

- Security layer for LLMs that strips PII and blocks prompt injections before reaching the model; fine-tuned RoBERTa-base on 160,000 examples across 10 PII classes to 0.940 out-of-distribution micro F1.
- Built a two-layer security agent with 18 regex injection filters; passed 22/22 integration and 34/34 feature tests.

Autonomous F1 Broadcast Director | Python, REST APIs, Pydantic

Feb. 2026

- Automated TV direction, enhancing race day viewing by rotating onboard camera feeds in MultiViewer; a Python daemon polls 9 OpenF1 endpoints every 4 seconds, scoring all 22 drivers across 18 weighted dimensions.
- Engineered switching logic with a 12-second dwell, 15% score threshold, 6-swap/minute budget, and Safety Car freezes.

TECHNICAL SKILLS

Languages: Python, Java, C, TypeScript, JavaScript, SQL

AI/ML & Data: LangChain, LangGraph, Anthropic API, OpenAI API, PyTorch, spaCy, OpenCV, Pydantic, matplotlib, pgvector, PostgreSQL, SQLite, Pandas, NumPy

Developer Tools: Git/GitHub, Docker, REST APIs, React, Next.js, Vercel, Railway

Certifications: Anthropic Academy: MCP, Agent Skills, Subagents, AI Fluency | AWS & DeepLearning.AI: Generative AI & LLMs | Google Cybersecurity: Foundations, Security Risks, Networks | University of Michigan: Programming in C